

1. Calibrate the Merton Model by finding the parameters that minimize the least squares distance between predicted option prices and actual market prices. Explore if this model is an improvement on Black-Scholes. (see e.g.
<https://www.sciencedirect.com/science/article/pii/B9780123846822000256>)
2. Simulate Delta Hedging applied to price processes other than usual Geometric Brownian Motion. Discuss its effectiveness also when transaction costs are added and when hedging can take place at different times scales, such as only hourly, only daily etc..
3. Implement the Euler Explicit Method for solving the PDE for option prices under the Schwartz mean reverting model (see course notes), given the parameters. Pay special attention to the boundary conditions. Compare with a Monte Carlo simulation.
4. Compare Monte Carlo and Quasi Monte Carlo for a European Option in the Merton Model (see e.g. https://en.wikipedia.org/wiki/Quasi-Monte_Carlo_methods_in_finance). Note that Matlab has commands for generating various Quasi-Random sequences: <https://it.mathworks.com/help/stats/generating-quasi-random-numbers.html>.
5. Implement the Euler Implicit method for an Asian Average Strike Put. Compare with a Monte Carlo Simulation.
6. Simulate the Variance-Gamma process. Implement a Monte Carlo simulation for the evaluation of a European option under this process. Explore variance reduction methods. Explore the effect of the parameters on the price.
7. Implement the Cheuk-Vorst tree algorithm for discrete lookback options, i.e., when the "observation dates" are a subset of the computation times used (see Currency lookback options and observation frequency: a binomial approach, by Cheuk and Vorst, page 181).